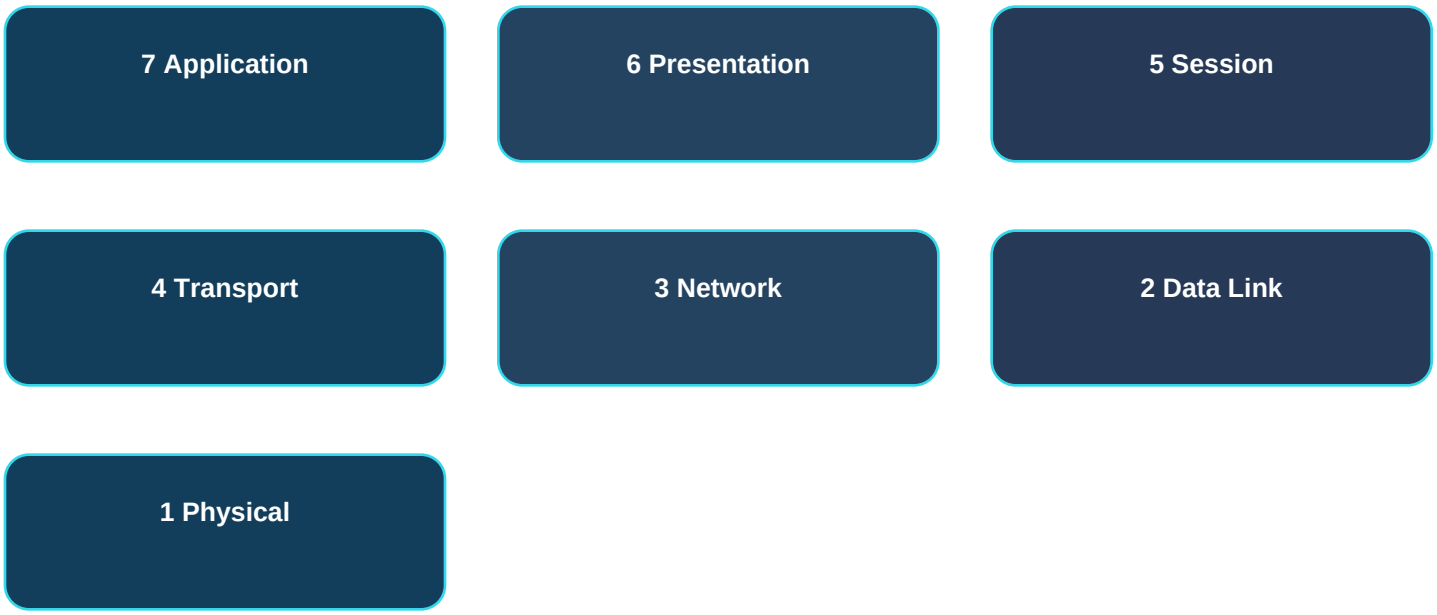
OSI Envelope Explorer

NET-OSI | Study Guide & Vocabulary | Created by Mr. Beaird

Visual Reference - System Components / Workflow



Preparation Checklist

- Visualize each OSI layer as an envelope around application data.
- At L4 add transport port details; at L3 add source/destination IP addresses.
- At L2 add local source/destination MAC addresses; L1 transmits bits.
- TLS encryption usually operates above TCP in the application stack; not exclusively at OSI L6.

Training reference diagram; not a vendor-specific photograph or configuration screenshot.

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Vocabulary and Technical Reference

Application (L7)
Application protocols such as HTTP and DNS.

Presentation (L6)
Data formatting, encoding, and related encryption concepts.

Session (L5)
Coordinates application sessions.

Transport (L4)
TCP/UDP ports, segmentation and delivery.

Network (L3)
IP packets, addressing, and routing.

Data Link (L2)
Ethernet frames and MAC addressing.

Physical (L1)
Bits, signals, cabling, and radio.

Review and Troubleshooting

[ ] What observable evidence confirms the objective was completed?

[ ] What two common configuration or hardware errors could cause failure?

[ ] Which tool, indicator, or command would you check first?

Student: _____ Date: _____